

$$f(\text{Abdul}) = 22 \quad f(\text{Brenda}) = 24 \quad f(\text{Carla}) = 21 \quad f(\text{Desiree}) = 22 \quad f(\text{Eddie}) = 24$$

$$f(\text{Felicia}) = 22$$

$$\text{Range: } \{22, 24, 21\} \quad \square$$

Example 3: Let f be the function that assigns the last two bits of a bit string of length 2 or greater to that string. For example, $f(11010) = 10$.

Domain: Set of all bit strings of length 2 or greater

Codomain and Range: $\{00, 01, 10, 11\}$. $\square$

Example 4: Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ assign the square of an integer to this integer. Then, $f(x) = x^2$, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, $\{0, 1, 4, 9, \dots\}$. $\square$

A function is called real-valued if its codomain is the set of real numbers, and it is called integer-valued if its codomain is the set of integers.

Two real-valued functions or two integer-valued functions with the same domain can be added, as well as multiplied.

Definition 3: Let f_1 and f_2 be functions from A to $\mathbb{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbb{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$(f_1 f_2)(x) = f_1(x) f_2(x)$$

Example 6: Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: By Definition 3, $f_1 + f_2 = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$

$$(f_1 f_2)(x) = f_1(x) \cdot f_2(x) = x^2(x - x^2) = x^3 - x^4 \quad \square$$

Definition 4: Let f be a function from A to B and let S be a subset of A . The image of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$. $f(S) = \{t \mid \exists s \in S (t = f(s))\}$. We also use the shorthand $\{f(s) \mid s \in S\}$ to denote this set.